Climate, fire and Singapore's haze, 2001–2026: a reproducible check with independent observations

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Abstract

The ocean state in June–August, published in September, signals how dry Indonesia's fire regions will be in August–October, how many fires will burn, and how hazy Singapore's air will be. It arrives before September and October, which held 67% of August–October fire detections in 2001–2025, but not before the season starts. I test this chain with independent public observations at every link, using an analysis plan fixed before the final tests.

All five primary tests hold after Holm correction for multiple testing. June–August ONI correlates with August–October rain over Kalimantan at r = −0.79 (95% interval −0.87 to −0.68; 1979–2025, n = 47). Beyond El Niño, each 1 °C of June–August DMI (the Indian Ocean Dipole index) goes with 1.7 mm/day less August–October rain over Sumatra (95% interval −2.7 to −0.6).

With the post-2015 policy period held fixed, each 1 mm/day less rain goes with about 1.55 times the fire detections in Sumatra and 1.44 times in Kalimantan (2001–2025, n = 25); at the same rain, fires since 2016 run about 77% and 47% lower with the break set at 2016. The size of that drop depends on the break year; the rain effect does not. Season fire counts rank-correlate with Singapore's August–October PM2.5 at ρ = 0.88 (95% interval 0.47 to 1.00; 2016–2025, n = 10).

Every planned robustness check keeps the sign and $p < 0.05$. The weakest link is the Dipole effect when model rain (ERA5) replaces observed rain ($p = 0.048$). The Singapore link rests on ten years and excludes the 2015 crisis. The results are correlational and were tested on data already explored; they say nothing about whether 2027 will be worse than 2026.

1 Introduction

Singapore's worst air comes from fires in Sumatra and Kalimantan, and much of the chain behind them is already known. This note re-checks that chain end to end with open data, up to the 2026 season, and adds the post-2015 policy shift that earlier simple correlations leave out.

What is known. Large fire years in Indonesia happen only when dry-season rain falls below a threshold (Field et al. 2009); fire and smoke respond sharply once rain stays under about 4 mm/day (Field et al. 2016). Seasonal rain forecasts can predict Kalimantan burnt area months ahead (Spessa et al. 2015). The 2019 positive Indian Ocean Dipole dried southern Sumatra and raised its fires (Iskandar et al. 2022). Model studies attribute most of Singapore's fire smoke to Sumatra, with Kalimantan giving about 14–15% in 2004–2009 (Reddington et al. 2014; Aouizerats et al. 2015).

The confounder. Since 2015, Indonesia's fire prevention has cut fire activity for a given drought, mainly on managed land (Sloan, Tacconi & Cattau 2021; Sloan, Gaveau et al. 2022). A simple rain–fire correlation across that break mixes weather with policy.

What this note does. It (1) uses June–August indices, which are published in September, ahead of the months when most fires burn; (2) uses a different, independent source at each link; (3) holds the post-2015 period fixed when linking rain to fires; and (4) fixes the tests and their correction in advance of the final run. Most of the chain is replication; the value is a reproducible, current, end-to-end check.

2 Data

Every link of the chain uses a different public source, so no two links share a data product (Table 1). All files were downloaded on 16 October 2026 and are stored unchanged with checksums.

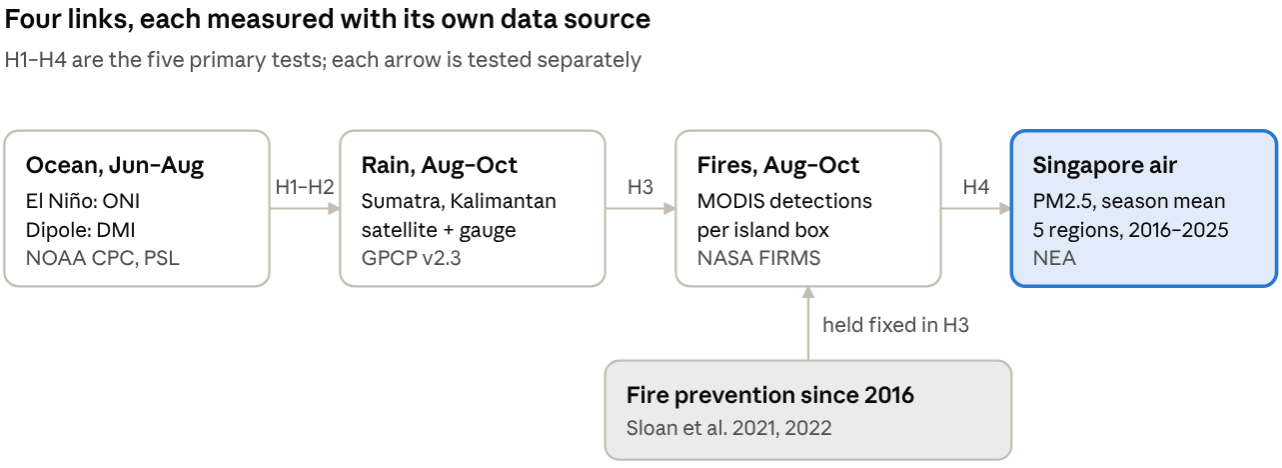


Figure 1 · the tested chain · 4 links, 5 primary tests, 1 controlled factor

The policy box is not a tested link: it is held fixed so that the rain–fire estimate is not confused with the post-2015 drop in fires.

Table 1. Data used

Link	Variable	Source	Years used	Licence
Ocean	ONI, June–August (El Niño index; ERSST v6)	NOAA CPC via NOAA PSL	1979–2025	US government data
Ocean	DMI, June–August (Indian Ocean Dipole; HadISST1.1, recent months preliminary)	NOAA PSL	1979–2025	Credit Met Office HadISST
Rain	August–October rain, mm/day; nearest 2.5° cell to 3 points per island	GPCP v2.3 monthly (satellite + gauge)	1979–2025	Cite Adler et al. 2003
Fires	August–October MODIS active-fire detections, all types, per island box	NASA FIRMS (standard product)	2001–2025	NASA open data
Singapore	August–October mean 24-h PM2.5, average of 5 regions	NEA via data.gov.sg	2016–2025	Singapore Open Data Licence
Checks only	ERA5 rain and pressure (SOI)	Open-Meteo historical API	1950–2026	CC BY 4.0

Island boxes: Sumatra 100–106.5°E, 5°S–1°N plus northern Riau (100–101.9°E, 1–2.3°N); Kalimantan 108.5–116.5°E, 4.5°S–1°N. The boxes exclude Peninsular Malaysia, Singapore and most of Sarawak; the Sumatra box includes Bangka island.

Known gaps, kept and not filled: NEA data start in 2016, so the 2015 crisis is outside the Singapore series. NEA has no data for 20 days in August–October, 13 of them in 2019, including peak haze days; 2019 haze is therefore understated. The 2026 season appears only as near-real-time fire data and partial-season Singapore data, outside the tests.

3 Methods

Five primary tests were written into a plan and frozen before the final run; all other analyses are labelled robustness or exploratory.

Honesty statement. The plan was written after four exploratory rounds had already looked at these data, including 2026. It is therefore not a pre-registration. It fixes the tests, definitions and correction so that results could not be tuned further. The only truly unseen data are future seasons.

Primary tests. Each year is one observation. Rain and fire variables are August–October; ocean indices are June–August. Those indices are published in September, so they overlap the first month of the season: they give warning ahead of September and October (67% of August–October detections in 2001–2025; August was the largest month in 8 of 25 years), not before the season starts.

1. **H1.** Pearson correlation of ONI with Kalimantan rain, 1979–2025.
2. **H2.** Sumatra rain regressed on ONI and DMI, 1979–2025; the test is on the DMI coefficient.
3. **H3a, H3b.** For each island, 2001–2025:

$$log_{10}(fires) = a + b \cdot rain + c \cdot post2016 + e$$

The test is on b; post2016 is 1 from 2016 onward and absorbs the policy shift.

4. **H4.** Spearman rank correlation of log10 fires (both islands) with Singapore's season-mean PM2.5, 2016–2025.

The five p-values are adjusted with the Holm method at α = 0.05.

Robustness checks (all planned in advance, all reported): leave-one-year-out for every test; a linear time trend instead of post2016; fires counted as vegetation type only; ERA5 rain instead of GPCP; Pearson instead of Spearman for H4; 95% intervals from 10,000 paired bootstrap resamples. A planned VIIRS cross-check was not run.

Logged deviations. A Durbin–Watson residual check was added and reported as a diagnostic only. Bootstrap resamples with a constant column were skipped (1 of 10,000 in one check); this rule was added after the first run had printed results, and changed no reported number. The main statistics were recomputed with separate code and matched to four decimals.

4 Results

All five primary tests reject at α = 0.05 after Holm correction (Table 2), and each link keeps its sign under every robustness check.

Table 2. Primary tests

Test	n	Estimate	95% bootstrap interval	Holm-adjusted p
H1 ONI → Kalimantan rain (Pearson r)	47	−0.786	−0.869 to −0.678	3.0×10^{-10}
H2 DMI → Sumatra rain, ONI fixed (mm/day per °C)	47	−1.674	−2.705 to −0.611	0.0030

Test	n	Estimate	95% bootstrap interval	Holm-adjusted p
H3a Rain → Sumatra log10 fires, post2016 fixed (per mm/day)	25	−0.189	−0.246 to −0.134	1.4×10^{-6}
H3b Rain → Kalimantan log10 fires, post2016 fixed (per mm/day)	25	−0.160	−0.195 to −0.137	1.9×10^{-9}
H4 Fires → Singapore PM2.5 (Spearman ρ)	10	0.879	0.472 to 1.000	0.0016

4.1 The ocean sets up the dry season

A warmer central Pacific in June–August goes with a drier Kalimantan in August–October (H1). Dropping any single year leaves r between −0.81 and −0.76. Over Sumatra, the Indian Ocean Dipole adds drying beyond El Niño (H2): a 1 °C higher DMI goes with 1.7 mm/day less rain.

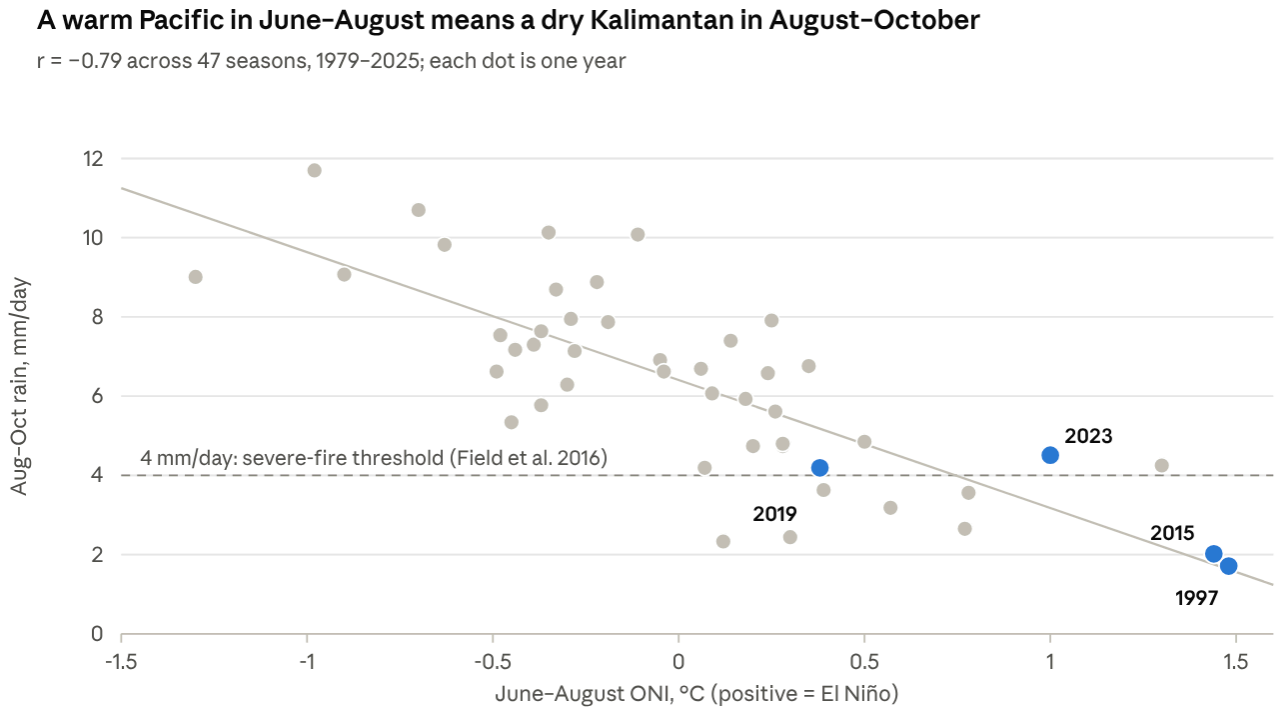


Figure 2 · GPCP v2.3 rain (nearest cells to 3 Kalimantan points); NOAA CPC ONI, ERSST v6 · 1979–2025, n = 47

Highlighted are the four years discussed in the text: 1997 and 2015 fall far below the 4 mm/day line, 2019 and 2023 just above it. The solid line is an ordinary least-squares fit.

4.2 Drier seasons burn more, but less than before 2016

On the log scale, the H3 coefficients mean each 1 mm/day less August–October rain multiplies fire detections by about 1.55 in Sumatra and 1.44 in Kalimantan. At the same rain, seasons since 2016 have about 0.23 times the detections in Sumatra (77% fewer, $p < 0.001$) and 0.53 times in Kalimantan (47% fewer, $p = 0.002$). The two models explain 82% and 89% of the year-to-year variance in log fires.

Those two percentages depend on putting the break at 2016, a year taken from the literature before the final tests. Moving the break anywhere from 2008 to 2021 leaves the rain coefficient almost unchanged (−0.18 to −0.23 in Sumatra, −0.16 to −0.18 in Kalimantan) but changes the drop: 0.21 to 0.51 times the detections in Sumatra and 0.53 to 0.82 times in Kalimantan, where the step is no longer significant for breaks from 2018 on (Section 4.5).

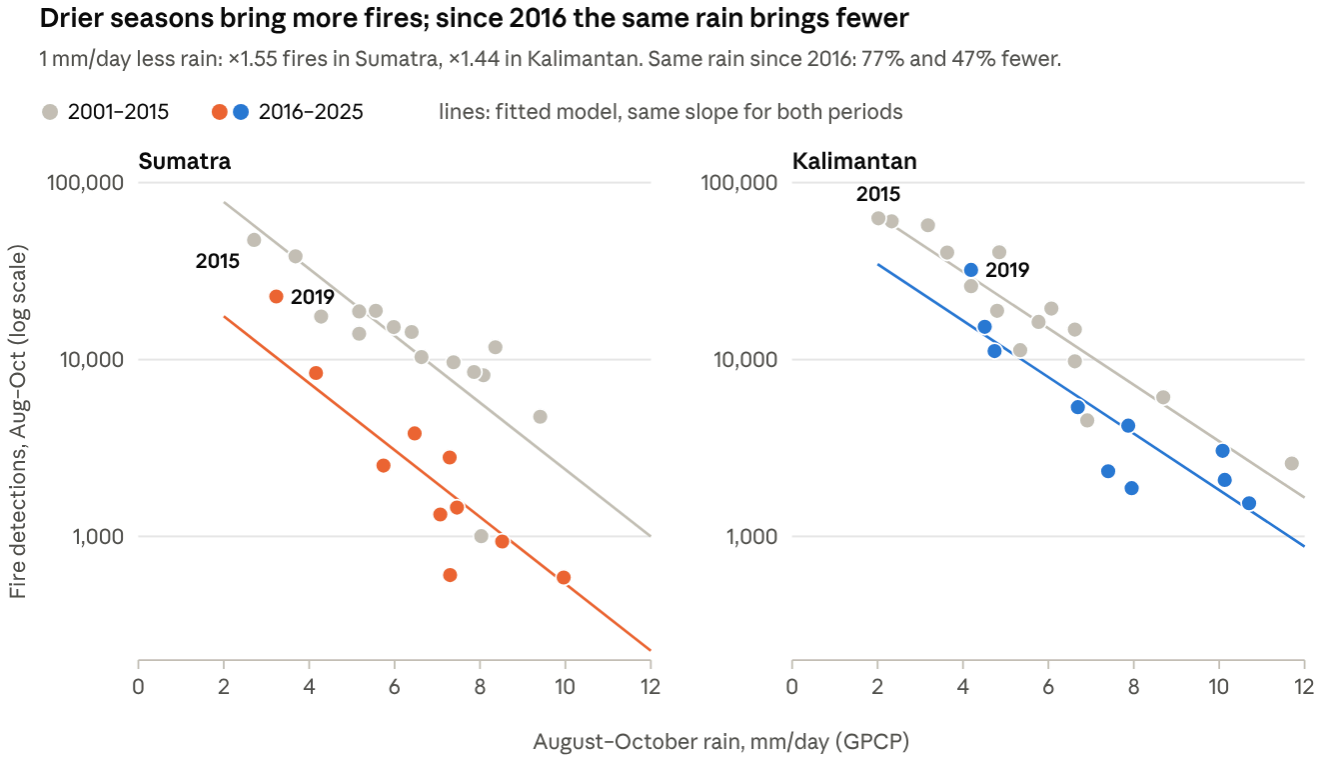


Figure 3 · NASA FIRMS MODIS detections (all types) and GPCP v2.3 rain · 2001–2025, n = 25 per island; lines from the H3 regressions

On both islands the coloured 2016–2025 line sits below the grey 2001–2015 line: the drought–fire link survives the policy shift, at a lower level.

4.3 More fires, hazier Singapore

Across 2016–2025, the order of seasons by fire count nearly matches their order by Singapore PM2.5 ($\rho = 0.88$). The interval is wide because there are only ten seasons. Leaving out 2019, the strongest season, still gives $\rho = 0.83$.

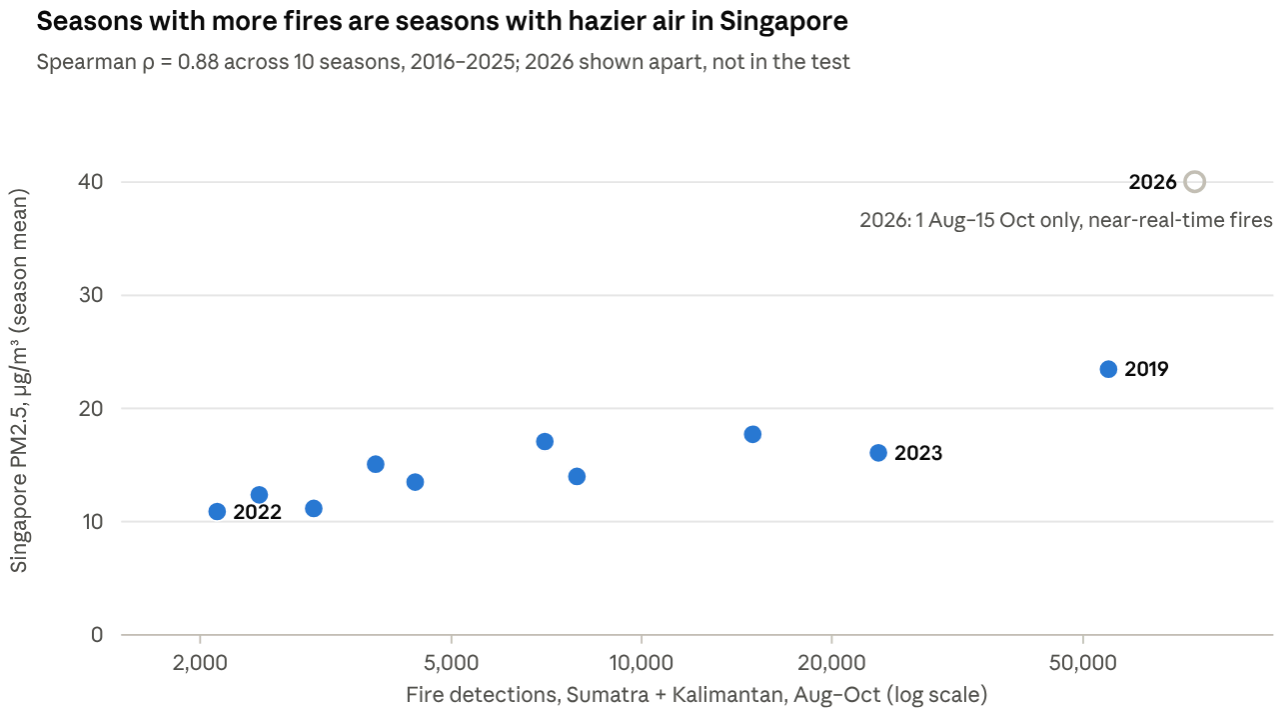


Figure 4 · NEA 24-h PM2.5 (5-region mean) and NASA FIRMS MODIS detections · Aug–Oct 2016–2025, n = 10; 2026 partial season

The open circle, 2026, continues the pattern, but it is a partial season with a different fire product and was not used in the test.

4.4 Robustness

Every check keeps each estimate's sign and $p < 0.05$. The closest call is H2 with ERA5 model rain ($p = 0.048$); with observed GPCP rain it is $p = 0.003$. Residuals of the Sumatra fire model show some carry-over from year to year (Durbin–Watson 1.44), so its p-value may be somewhat optimistic; its bootstrap interval still excludes zero.

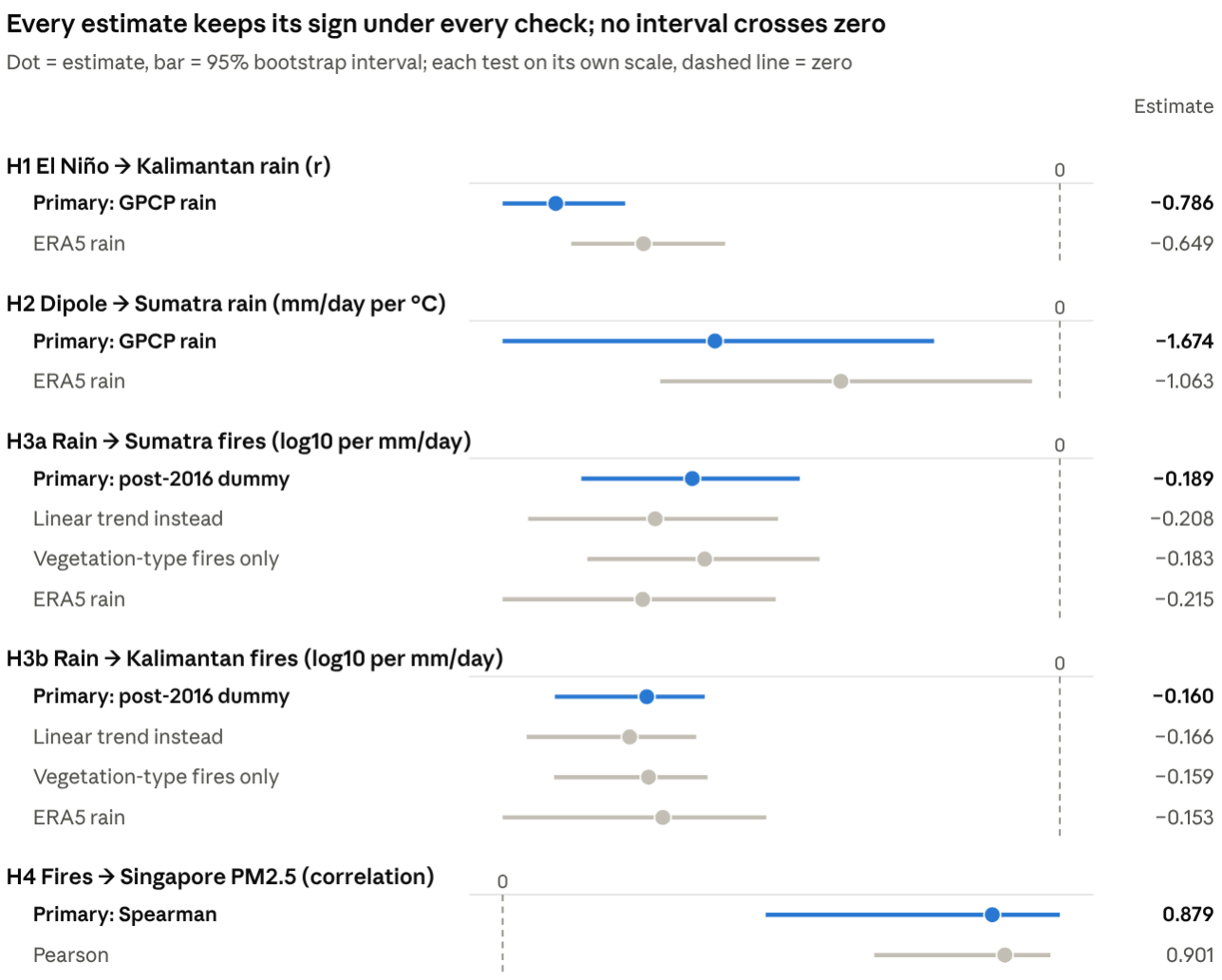


Figure 5 · results/confirmatory_primary.csv and confirmatory_robustness.csv · 10,000 paired bootstrap resamples, seed 20261016

The one interval that comes close to zero is H2 with ERA5 rain; the planned VIIRS fire check was not run.

4.5 Checks added after review

Two readers on the r/AskStatistics forum reviewed the design after the first version of this note. The checks below answer their points. They were not in the plan, so they are reported as post hoc, and none of them changes a primary result. The numbers are in `results/posthoc_checks.csv` and `results/posthoc_break_scan.csv`.

- **Count model.** No season has zero detections (the smallest is 586, in Sumatra). A negative binomial model with a log link gives 1.52 and 1.44 times the fires per 1 mm/day less rain, against 1.55 and 1.44 from the log-linear model.
- **Shared trends.** Fire counts look non-stationary, which fits the policy break. In year-to-year differences every link still holds: r = −0.84 for ONI and Kalimantan rain (n = 46), rain coefficients of −0.19 and −0.17 on log fires (n = 24), and ρ = 0.90 for fires and

Singapore PM2.5 (n = 9).

- **Break year.** The rain effect is stable for any break from 2008 to 2021; the size of the drop is not (Section 4.2).
- **Step or trend.** With a linear trend and the 2016 step in the same model, the trend adds nothing (p 0.73 and 0.72) while the step remains (p 0.002 and 0.04). Twenty-five years cannot fully separate the two.
- **Dependence between years.** A moving-block bootstrap with blocks of two to four years gives intervals for the rain coefficient of about −0.23 to −0.13 (Sumatra) and −0.20 to −0.14 (Kalimantan), close to the year-by-year bootstrap.

5 Exploratory findings

These observations were not tested and carry no significance claim; they point to what to test next.

Why 2019 was worse than 2023. 2023 had the stronger El Niño (June–August ONI 1.00 against 0.38), yet Singapore's August–October PM2.5 was lower (16.1 against 23.5 μg/m³). In 2019 Sumatra was drier (GPCP 3.23 against 4.16 mm/day) and had 2.7 times the fire detections (22,720 against 8,394). This fits the H2 result and the published link between the 2019 Dipole and southern Sumatra fires. But the 2023 Dipole was even stronger (August–October DMI 0.86 against 0.76), so the Dipole alone does not explain the difference.

2026 does not fit the Sumatra-first view. From 1 August to 10 October 2026, Kalimantan had 53,502 near-real-time detections and Sumatra 15,386, so Sumatra held only about 22% of fire detections. Singapore's mean PM2.5 over those dates was 36.0 μg/m³, the highest since the NEA series began in 2016. Model studies credit most of Singapore's smoke to Sumatra, so either Kalimantan smoke reached Singapore in unusual amounts, or Sumatra's fewer fires were unusually smoky (peat). Wind and smoke-transport data would separate the two; they are not analysed here.

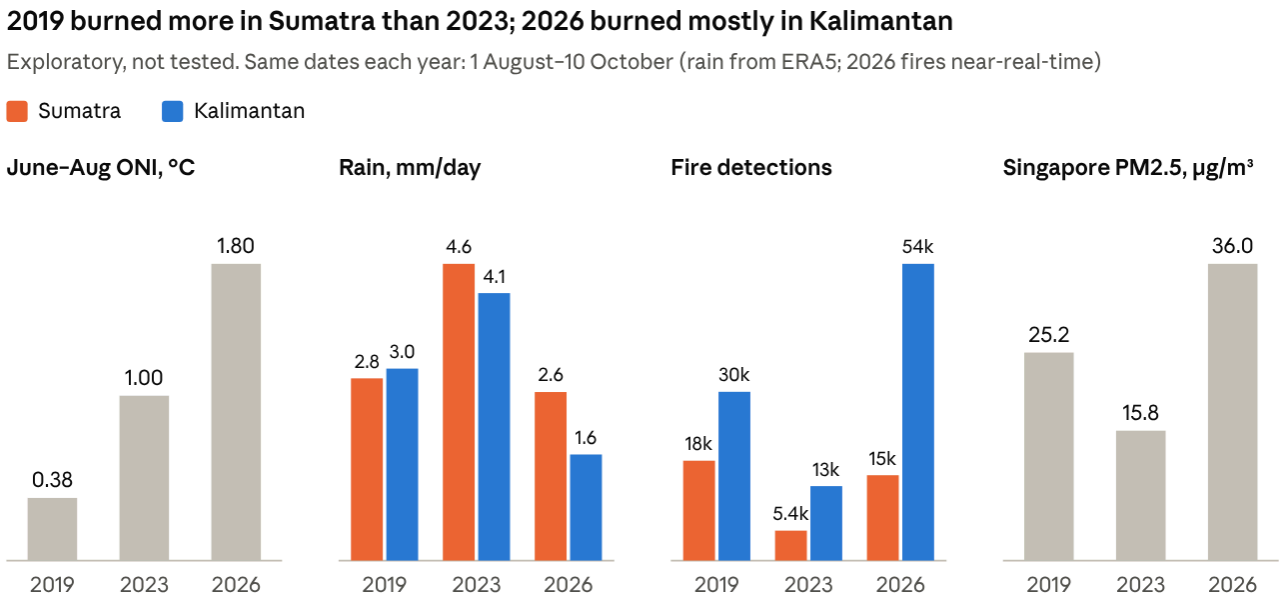


Figure 6 · NOAA CPC ONI; ERA5 rain via Open-Meteo; NASA FIRMS MODIS (2026 near-real-time); NEA PM2.5 · 1 Aug–10 Oct of each year

The same dates are used for all three years so that 2026 can be compared; GPCP rain is not yet available for 2026, so this figure uses ERA5.

Satellite labelling may hide peat fires. FIRMS flags some detections as "other static land source". In August–October 2015, 13,963 Sumatra detections carried that label, against 9 in 2019, clustered in South Sumatra peatland. They look like long-burning peat fires caught by a mask meant for industrial sites, as Sofan et al. 2022 warn for VIIRS. This note therefore counts all detections; counting vegetation-type only does not change any result (Section 4.4).

6 Discussion and limitations

The practical message is a short-lead warning: once the June–August ocean indices are published in September, they indicate whether the rest of the season is likely to be hazy. A longer lead would need earlier indices (for example May–July), which this note did not test.

What can be said.

- The chain ocean → rain → fires → Singapore PM2.5 holds at every link with independent observations, through 2025.
- Fire prevention since 2015 has lowered fires for any given rainfall, but has not broken the link between drought and fire.
- The Dipole matters for Sumatra beyond El Niño, and Sumatra matters most for Singapore in model studies; both indices belong in a warning.

What cannot be said.

- **Nothing here forecasts 2027.** The warning signal is the June–August state of the same year. Current forecasts (WMO, August 2026) expect this El Niño to peak at the end of 2026 and last into early 2027, but say nothing about mid-2027.
- **Correlation, not proof of cause.** The chain is physically plausible and matches the literature, but yearly correlations cannot rule out other shared drivers.

Limits.

1. The Singapore link has ten seasons and no 2015; its interval is wide (ρ from 0.47 to 1.00).
2. The tests were run on data already explored (Section 3).
3. Fire counts measure detections, not smoke; peat fires emit far more per detection. The MODIS satellites' orbits have drifted in recent years; this is not corrected.
4. GPCP cells are 2.5° and partly over sea; 2019 Singapore haze is understated by missing days.
5. Sumatra fire residuals carry over between years, so that model's p-value may be optimistic.
6. The size of the post-2015 drop depends on the break year (Section 4.5); the 77% and 47% figures hold only for a 2016 break.
7. With 25 years, a one-off drop and a gradual decline cannot be fully told apart.

Next. The true out-of-sample test is the 2027 season: record the June–August 2027 indices and the fire count, write down the expected Singapore PM2.5, then compare. Smoke transport, fire radiative power and VIIRS detections are the next analyses; palm-oil economics is a separate study.

7 Data, code and references

All raw files, code, the frozen plan (`docs/research_plan_v1.md`) and every result table are in the repository Shelikeit69/haze-drivers-research. Running the scripts listed in the README, in order, reproduces every number here.

Acknowledgements. Two commenters on r/AskStatistics raised the points answered in Section 4.5. This version, revised on 19 October 2026, adds that section and the break-year caveats.

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中文摘要

6–8 月的海洋状态（9 月公布）可以提示当年 8–10 月印尼火区有多干、火有多少、新加坡空气有多差。它能在 9–10 月之前给出预警（2001–2025 年 8–10 月的火点有 67% 出现在 9–10 月），但不是在雾霾季开始之前。本文用每一环都相互独立的公开观测数据检验这条链，检验方法在最终运行前已经固定。

5 项主检验经 Holm 多重比较校正后全部显著。6–8 月 ONI 与加里曼丹 8–10 月降雨的相关系数为 r = −0.79（1979–2025 年，共 47 年）。在 El Niño 之外，6–8 月印度洋偶极子指数（DMI）每高 1°C，苏门答腊 8–10 月降雨约少 1.7 毫米/日。

控制 2016 年后的政策变化后，降雨每少 1 毫米/日，苏门答腊火点约增至 1.55 倍，加里曼丹约 1.44 倍；以 2016 年为断点，同样雨量下 2016 年后两岛火点分别少约 77% 和 47%；降幅会随断点年份变，降雨效应基本不变。火点数量与新加坡 8–10 月 PM2.5 的秩相关为 ρ = 0.88（2016–2025 年，共 10 年）。

换数据源、逐年剔除、换统计方法之后，所有结论方向不变，p 值都低于 0.05。最弱的一环是用 ERA5 模型降雨检验偶极子作用，p = 0.048。

另有两个未经检验的观察。2023 年的 El Niño 比 2019 年强，但 2019 年苏门答腊更干，火点约为 2023 年的 2.7 倍，新加坡烟霾也更重。2026 年约 78% 的火点在加里曼丹，新加坡 PM2.5 却是 2016 年以来最高，这和"新加坡的烟主要来自苏门答腊"的模型研究对不上。

局限也要说清楚：新加坡这一环只有 10 年数据，不含 2015 年；检验用的数据此前做过探索性分析；结果只是相关，不能证明因果。本文也不预测 2027 年的雾霾会比 2026 年更重或更轻。另外，火点降幅取决于断点放在哪一年，25 年的数据也分不清这是一次性下降还是逐年下降。